7.3 SegNet

- the idea of the SegNet model is to perform step-wise upsampling
- SegNet has an encoder network and a corresponding decoder network, followed by a final pixelwise classification layer
- this architecture is illustrated in Figure 7
- the encoder network consists of 13 convolutional layers which correspond to the first 13 convolutional layers in the VGG-16 network
- we can therefore initialize the training process from weights trained for classification on large datasets, e.g., ImageNet
- the fully connected layers are discarded in favor of retaining higher-resolution feature maps at the deepest encoder output
- each encoder layer has a corresponding decoder layer and hence the decoder network has 13 layers
- the final decoder output is fed to a multi-class softmax classifier to produce class probabilities for each pixel independently

7.3 SegNet

- each encoder in the encoder network performs convolution to produce a set of feature maps
- these are then batch normalized, and then an elementwise ReLU is applied
- following that, max-pooling with a 2×2 window and stride 2 is performed and the resulting output is sub-sampled by a factor of 2
- while several layers of max-pooling can achieve more translation invariance for robust classification, correspondingly there is a loss of spatial resolution of the feature maps
- the increasingly lossy (boundary detail) image representation is not beneficial for segmentation, where boundary delineation is vital
- therefore, it is necessary to capture and store boundary information in the encoder feature maps before sub-sampling is performed
- because all the encoder feature maps (after subsampling) cannot be stored, a more efficient way to store this information involves storing only the max-pooling indices, i.e., the locations of the maximum feature value in each pooling window is memorized for each encoder feature map

7.3 SegNet

- the appropriate decoder in the decoder network upsamples its input feature maps using the memorized encoder max-pooling indices from the corresponding encoder feature maps, i.e., it performs max-unpooling
- this step produces sparse feature maps
- these feature maps are then processed using trainable convolutional layers to produce dense feature maps, i.e., the goal of the convolutional layers is to refine the upsampling
- a batch normalization step is then applied to each of these maps
- note that the decoder corresponding to the first encoder (closest to the input image) produces a multi-channel feature map, although its encoder input has 3 channels (RGB)
- this is unlike the other decoders in the network which produce feature maps with the same number of size and channels as their encoder inputs
- the high-dimensional feature representation at the output of the final decoder is fed to a trainable softmax classifier

7.3 SegNet

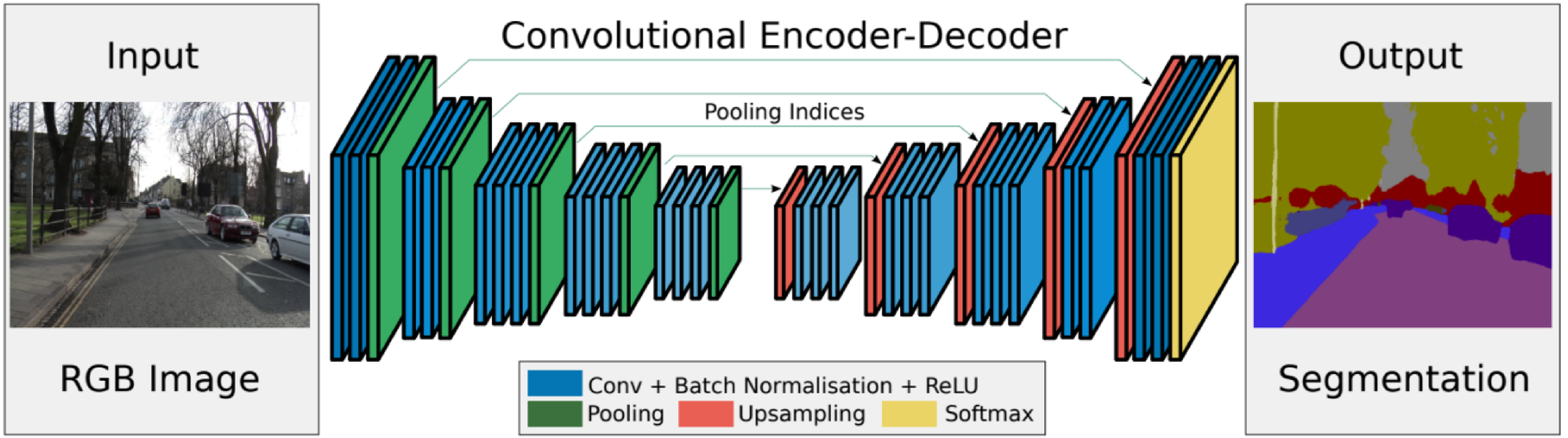


Figure 7: An illustration of the SegNet architecture. There are no fully connected layers and hence it is only convolutional. The decoder upsamples its input using the transferred pooling indices from the encoder to produce a sparse feature map. It then performs convolution with trainable filters to densify the feature map. The final decoder output feature maps are fed to a softmax classifier for pixelwise classification.